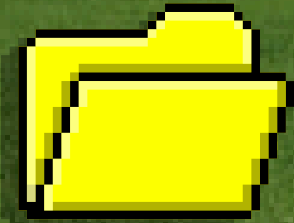


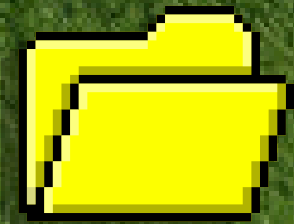
(01) Data



(02) Features



(03) Outcomes



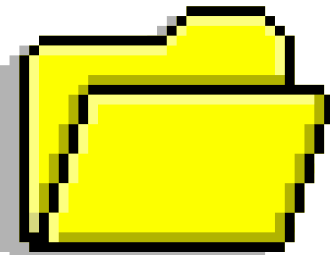
(04) What's Next?

Problematic Internet Usage in Youths

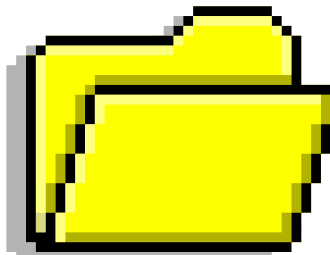
an Audit of an ADS

Luke Sullivan and Ashley Tewari

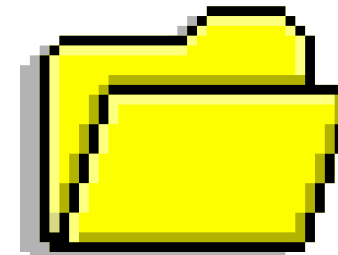
Today's Agenda



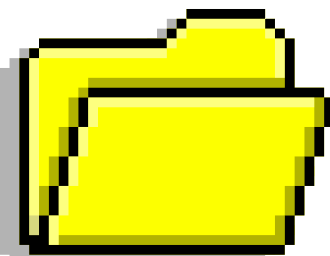
(01) Background



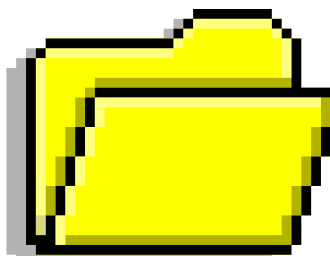
(02) Inputs



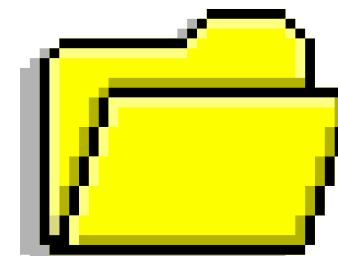
(03) Outputs



(04) Outcomes

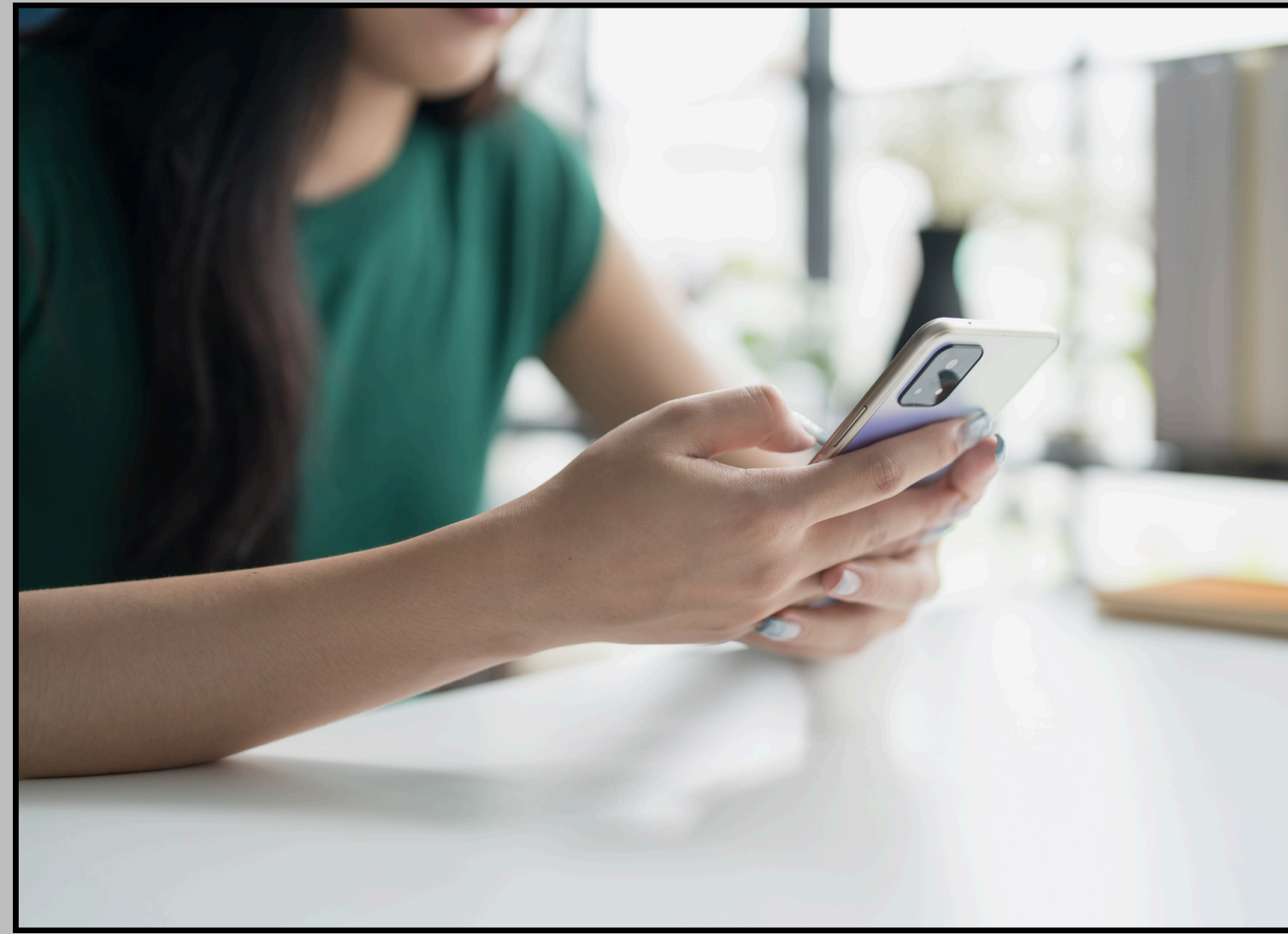


(05) Summary



(06) What's Next?

Background



Internet addiction?

The internet has fundamentally changed how people interact with the world.

Diagnostic models that predict mental health and behavioral disorders are ethically paramount.

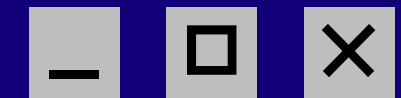


The Data

**Child Mind
Institute**

- Healthy Brain Network
- 5,000+ children, aged 5-21
- clinically-referred sample
- #1 Kaggle Solution

Inputs



➤ Physical Measurements

- height, weight, BMI, blood pressure, heart rate

➤ Fitness Assessments

- grip strength, curl-ups, push-ups, trunk lift, endurance

➤ Body Composition (BIA)

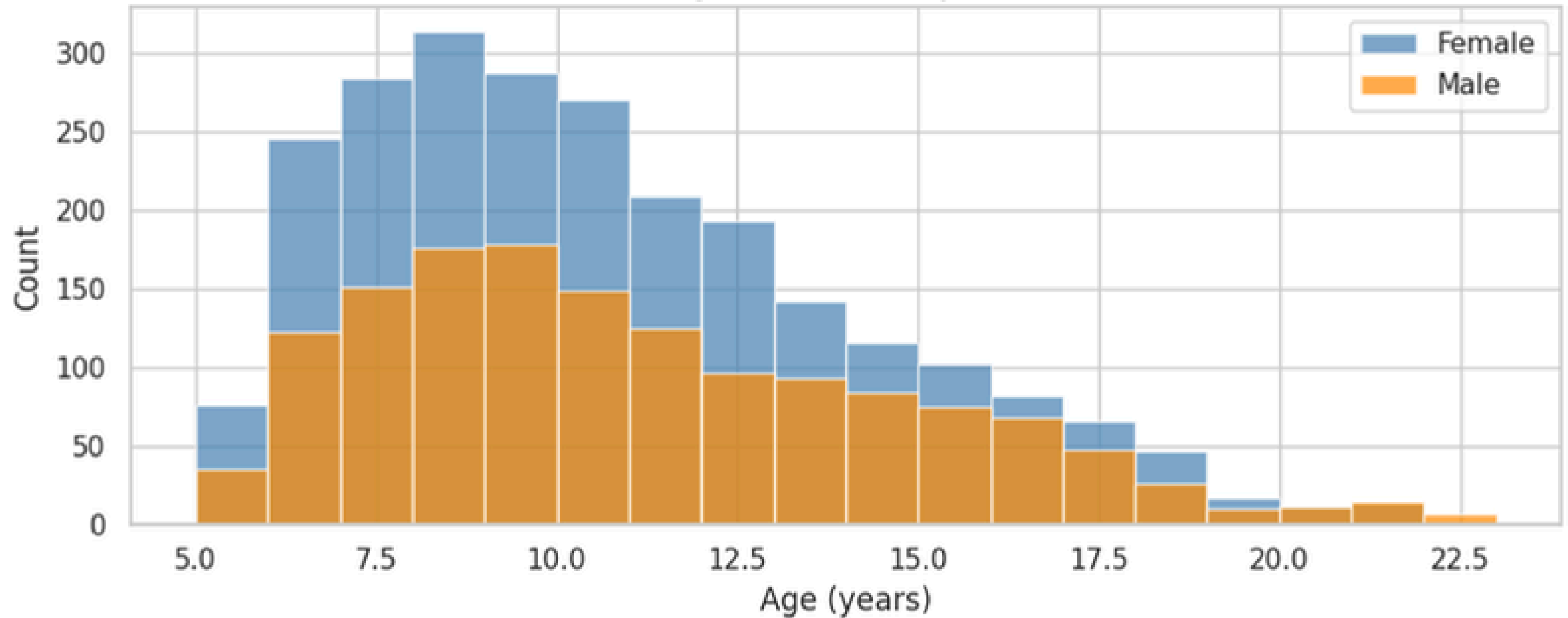
- fat percentage, muscle mass, metabolic rate

➤ Feature Correlation

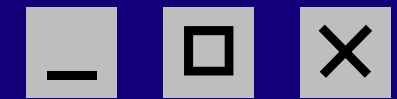
- BIA features are highly correlated (0.48)
- actigraphy and questionnaires are independent

➤ Missing Data!

Age Distribution by Sex



Outputs



➤ Severity Impairment Index

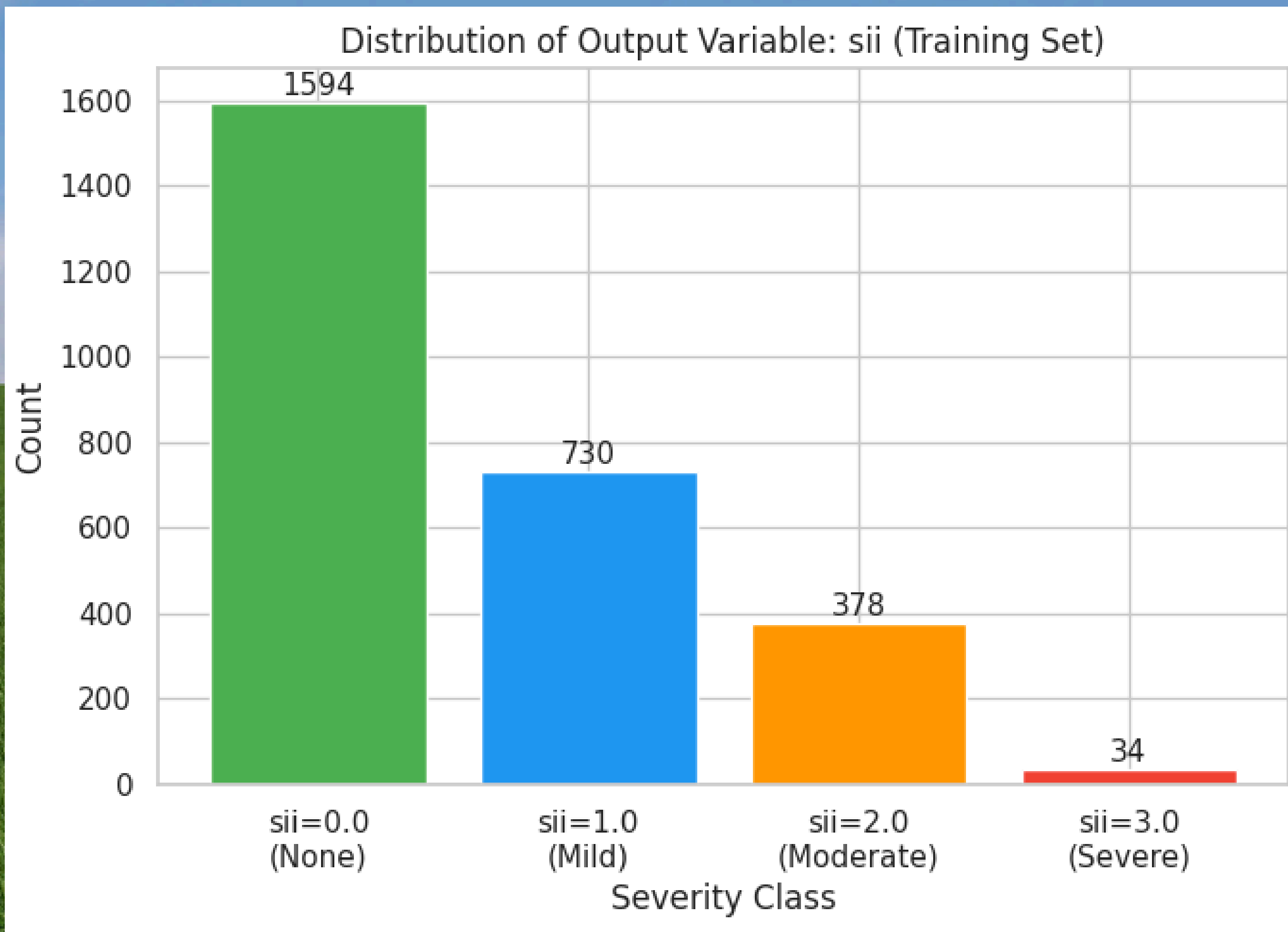
- Severity scale from 0 to 3, predicting problematic internet usage

➤ Class Distribution

- 1594 None cases vs 34 Severe cases, imbalance

➤ Evaluation Metric

- Quadratic Weighted Cohen's Kappa (QWK) = 0.47

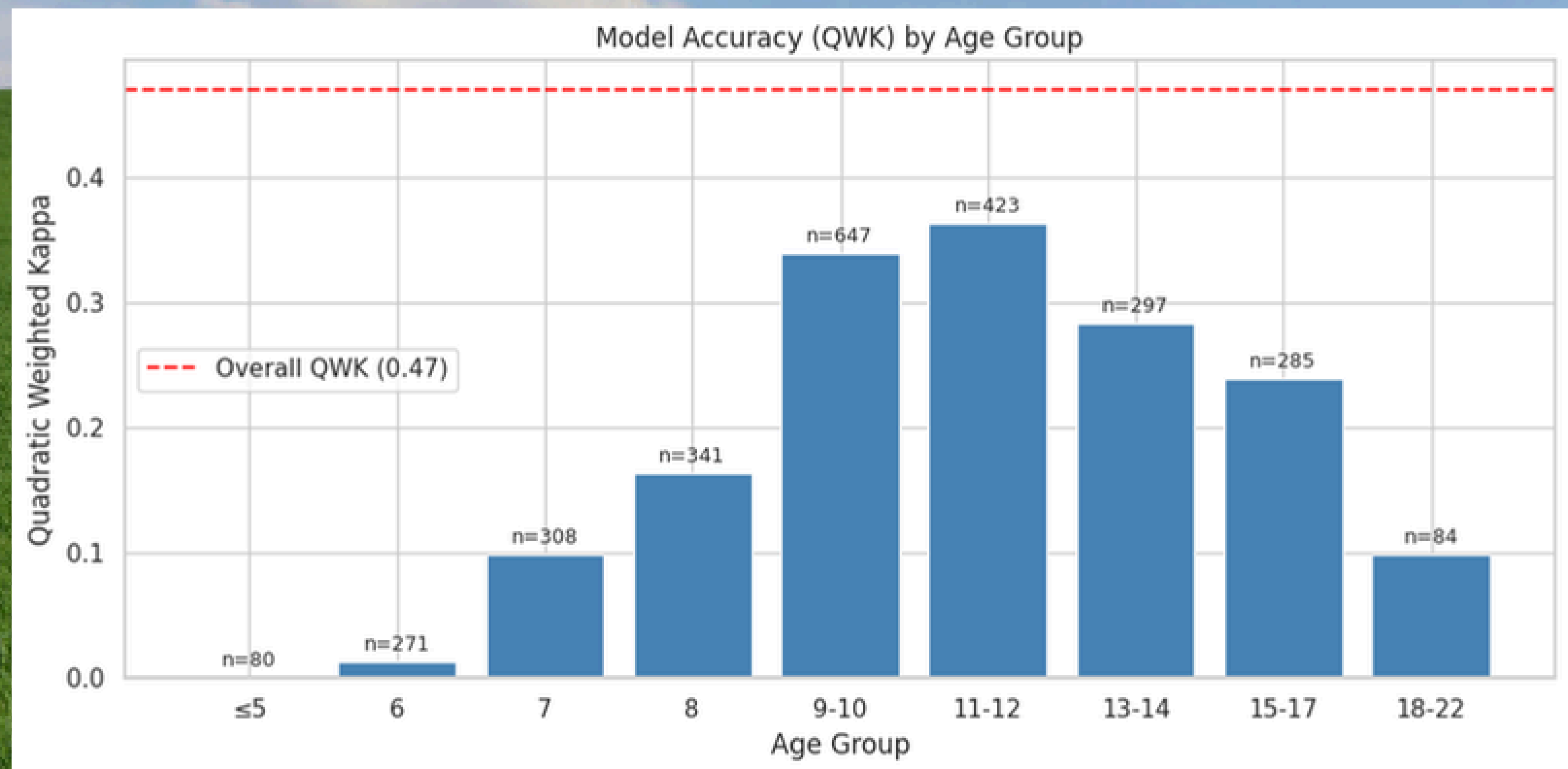


Outcomes

(01)

Accuracy

- QWK varies greatly by age group
- no age group reaches 0.47



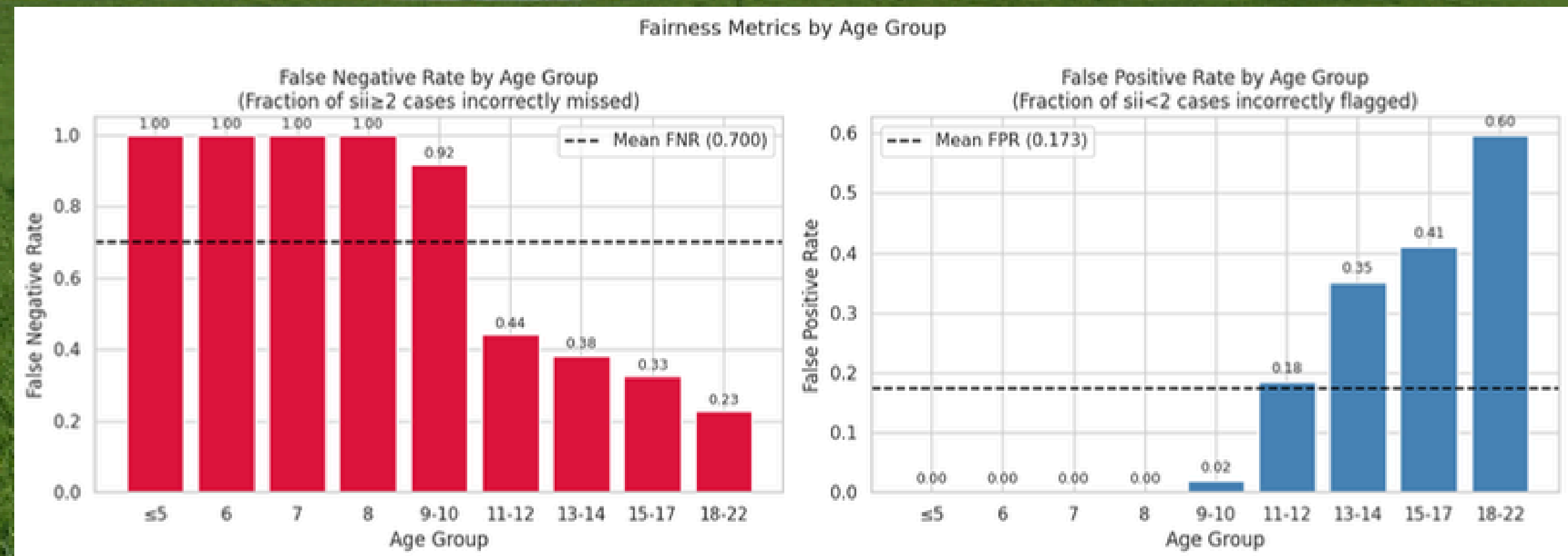
Outcomes

(02)

Fairness

➤ FNR = 1.0 for
children from ages 5
to 8

➤ the model misses
every severe case

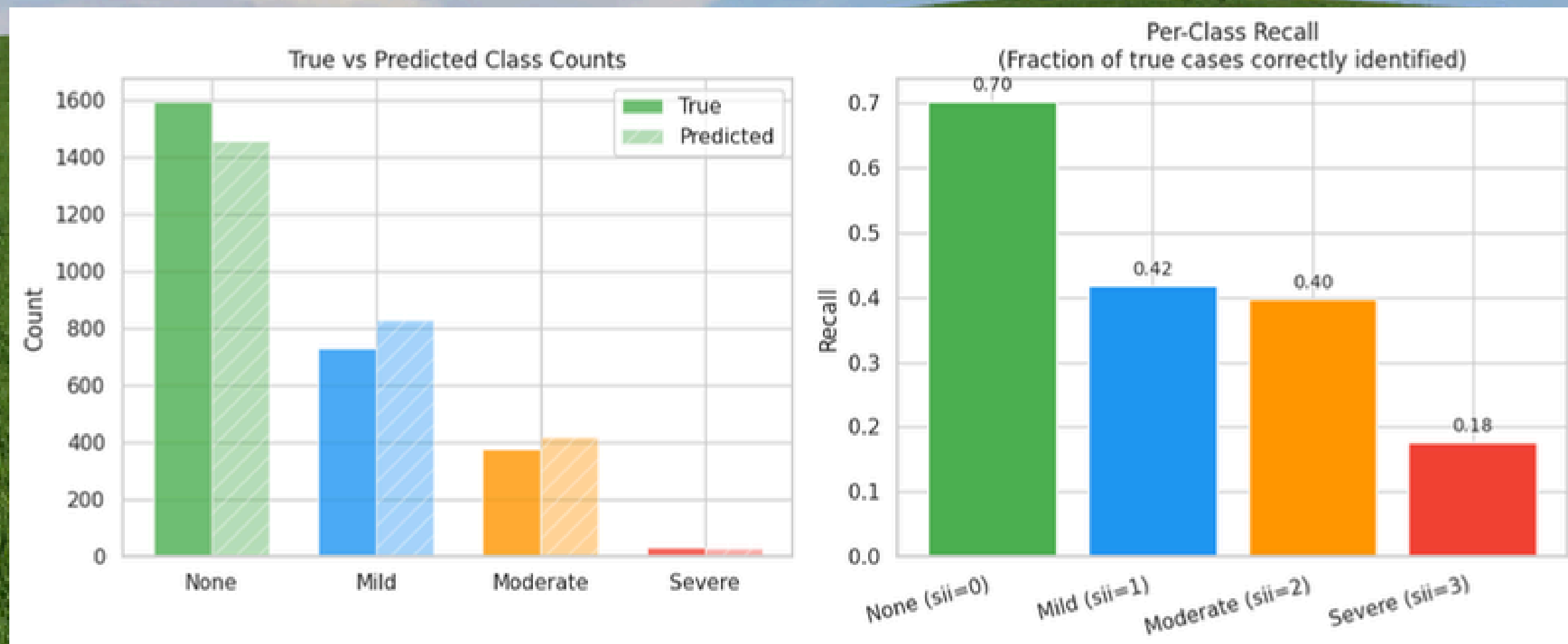


Outcomes

(03)

Robustness

- the model is stable across seeds
- $\text{STD} = 0.007$
- still fails on rare and severe cases
- $\text{recall} = 0.176$





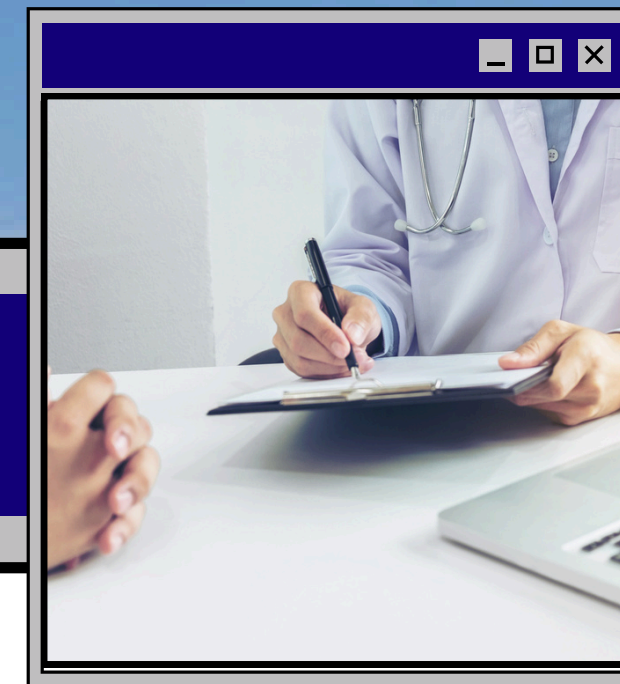
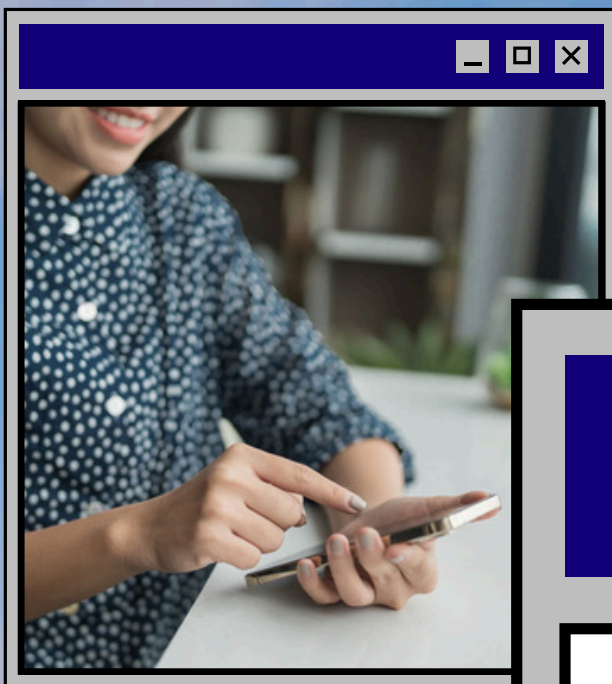
Summary

- Our model achieves $QWK = 0.47$ but is unable to account for young children with severe cases.
- There is an FNR of 1.0 for children under 8 years old.
- The model is stable but has a recall of 18% for severe cases.
- It is not clinically capable or ethically deployable given the outcomes.

What's Next?

- (01) train models per age band
- (02) oversample severe cases
- (03) flag imputed values





Thank you!

